



SECI1143 PROBABILITY AND STATISTICAL DATA ANALYSIS

SEMESTER 2 (2023/2024)

SECTION 03

ASSIGNMENT 3

Chapter 5: Hypothesis Testing &

Chapter 6: Chi-Square Test and Contingency Analysis

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Question 1

$$a) \sum x_i = 345 + 420 + 390 + 410 + 380 + 400 + 430 + 360 + 375 + 390 + 405 + 415 + 400 + 385 + 395$$

$$= 5900$$

$$\hat{p} = \frac{5900}{15}$$

$$= 393.33$$

b) Confidence interval = 95%

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

$(x_i - \bar{x})$:

$$(345 - 393.33)^2 = 2335.789$$

$$(420 - 393.33)^2 = 711.289$$

$$(390 - 393.33)^2 = 11.089$$

$$(410 - 393.33)^2 = 277.889$$

$$(380 - 393.33)^2 = 177.689$$

$$(400 - 393.33)^2 = 44.489$$

$$(430 - 393.33)^2 = 1344.689$$

$$(360 - 393.33)^2 = 1110.889$$

$$(375 - 393.33)^2 = 335.989$$

$$(390 - 393.33)^2 = 11.089$$

$$(405 - 393.33)^2 = 136.189$$

$$(415 - 393.33)^2 = 469.589$$

$$(400 - 393.33)^2 = 44.489$$

$$(385 - 393.33)^2 = 69.389$$

$$(395 - 393.33)^2 = 2.789$$

$$\sum (x_i - \bar{x})^2 = 7083.335$$

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

$$= \sqrt{\frac{7083.335}{15-1}}$$

$$= 22.5$$

Critical value : $t_{\alpha/2, n-1}$

$$: t_{0.025, 14} \approx 2.145$$

$$\bar{x} \pm (t \text{ critical value}) \left(\frac{s}{\sqrt{n}} \right)$$

$$393.33 \pm (2.145) \left(\frac{22.5}{\sqrt{15}} \right)$$

$$= (380.87, 405.80)$$

Conclusion :

We are 95% confident that the average monthly expenditure on groceries for the demographic group lies between RM380.87 and RM405.80.

Question 2

$$(a) H_0: \mu \leq 5$$

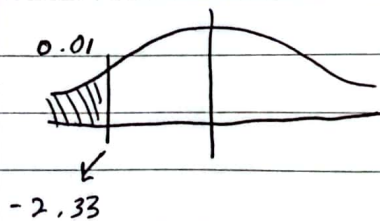
$$H_1: \mu < 5$$

$$(b) z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

$$= \frac{4.6 - 5}{1.2 / \sqrt{500}}$$

$$= -7.45$$

$$(c) \alpha = 0.01$$



$$0.5 - 0.01 = 0.49$$

(d) Conclusion: H_0 is rejected. There is sufficient evidence to conclude that $\mu < 5$. This is because, the value of calculated z is smaller than the value of significance level.

(e) The bank manager now can determine that the average waiting time for customers in the bank's queue is less than 5 minutes.

Question 3

$$n = 25$$

$$\alpha = 0.05$$

$$\bar{x} = 183$$

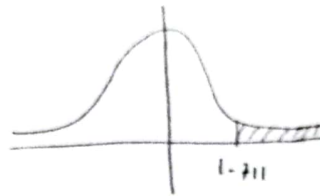
$$s = 12$$

$$H_0: \mu = 160$$

$$H_1: \mu > 160$$

$$\begin{aligned} t &= \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} \\ &= \frac{183 - 160}{\frac{12}{\sqrt{25}}} \\ &= 9.58 \end{aligned}$$

$$t_{0.05, 24} = 1.711$$



$\therefore$ since $t_0 = 9.58 > 1.711$, we reject the $H_0: \mu = 160$ at the 0.05 level of significance. There is sufficient evidence to support the claim mean score for the students from this university is greater than 160.

Question 4

$$\mu_0 = 50 \quad n = 38 \quad \bar{x} = 51 \text{ cm} \quad \alpha = 0.01 \quad s = 3.6$$

$$H_0 : \mu = 50$$

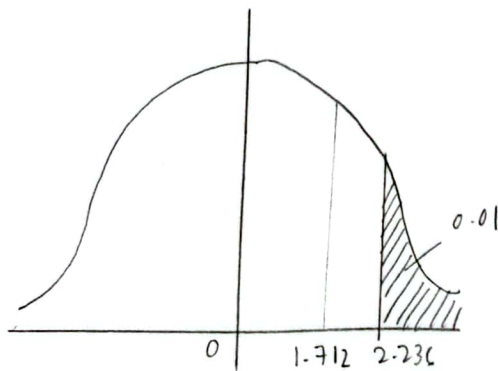
$$H_1 : \mu > 50$$

$$n > 30$$

$$z = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$$

$$= \frac{51 - 50}{\frac{3.6}{\sqrt{38}}}$$

$$= 1.712$$



From table,

$$\alpha = 0.01$$

Critical value, t is 2.236

$$t_0 < t_{0.01, 38}$$

$$1.712 < 2.236$$

Conclusion:

Fail to reject H_0 .

There is not enough evidence to support the claim that the population mean length of steel bars is greater than 50 cm.

Question 5

$$n = 51$$

$$\mu = 16576$$

$$\sigma = 781$$

$$\alpha = 0.01$$

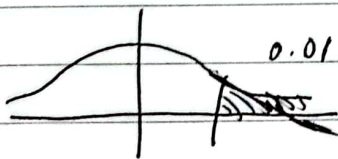
$$\text{female } \sigma = 7460$$

$$H_0: \sigma = 7460$$

$$H_1: \sigma > 7460$$

$$\begin{aligned} \chi^2 &= \frac{(n-1)s^2}{\sigma^2} \\ &= \frac{50(7871)^2}{7460^2} \\ &= 55.66 \end{aligned}$$

$$\begin{aligned} \text{Degree of freedom} &= n-1 \\ &= 51-1 \\ &= 50 \end{aligned}$$



$$\text{Critical } \chi^2 \text{ value} = 76.154$$

Conclusion: The value of critical value is higher than the value of calculated critical value. Hence, H_0 is failed to reject because males have standard deviation that is not greater than the standard deviation of 7460 words for females.

Question 6

$$a) \text{ Female, } \bar{X}_1 = \frac{4.3 + 4.3 + 4.4 + 4.0 + 3.4 + 4.7 + 2.9 + 4.0 + 4.3 + 3.4 + 3.4 + 3.3}{12}$$

$$= \frac{46.4}{12}$$

$$= 3.87$$

$$S_1^2 = \frac{(4.3 - 3.87)^2 + (4.3 - 3.87)^2 + (4.4 - 3.87)^2 + (4.0 - 3.87)^2 + (3.4 - 3.87)^2 + (4.7 - 3.87)^2 + (2.9 - 3.87)^2 + (4.0 - 3.87)^2 + (4.3 - 3.87)^2 + (3.4 - 3.87)^2 + (3.4 - 3.87)^2 + (3.3 - 3.87)^2}{11}$$
$$= 0.3170$$

$$S_1 = \sqrt{0.3170}$$
$$= 0.5630$$

$$\text{Male, } \bar{X}_2 = \frac{4.5 + 3.7 + 4.2 + 3.9 + 3.1 + 4.0 + 3.8 + 3.4 + 4.5 + 3.8 + 4.3 + 4.4 + 4.1 + 4.2 + 4.0 + 4.4}{16}$$

$$= 4.02$$

$$S_2^2 = \frac{(4.5 - 4.02)^2 + (3.7 - 4.02)^2 + (4.2 - 4.02)^2 + (3.9 - 4.02)^2 + (3.1 - 4.02)^2 + (4.0 - 4.02)^2 + (3.8 - 4.02)^2 + (3.4 - 4.02)^2 + (4.5 - 4.02)^2 + (3.8 - 4.02)^2 + (4.3 - 4.02)^2 + (4.4 - 4.02)^2 + (4.1 - 4.02)^2 + (4.2 - 4.02)^2 + (4.0 - 4.02)^2 + (4.4 - 4.02)^2}{15}$$

$$= 0.1563$$

$$S_2 = \sqrt{0.1563}$$
$$= 0.3953$$

b) $\alpha = 0.05$

$H_0: \bar{X}_1 = \bar{X}_2$

$H_1: \bar{X}_1 \neq \bar{X}_2$

$\bar{X}_1 = 3.87$ $\bar{X}_2 = 4.02$
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$$t_0 = \frac{\bar{X}_1 - \bar{X}_2 - A_0}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{3.87 - 4.02 - 0}{0.47 \sqrt{\frac{1}{12} + \frac{1}{16}}} = -0.84$$

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

$$= \frac{(12 - 1)0.3170 + (16 - 1)0.1563}{12 + 16 - 2}$$

$$= 0.22$$

$$S_p = \sqrt{0.22}$$

$$= 0.47$$

c) Since $-2.056 < -0.84 < 2.056$, H_0 is fail to reject, that is at the level of 0.05 level of significance. we do not have strong evidence to conclude that the two samples have different mean.

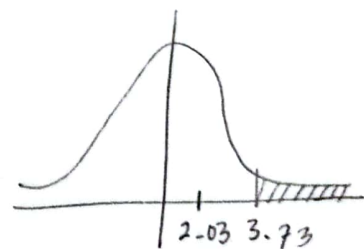
d) $H_0: \sigma_2^2 = \sigma_1^2$

$H_1: \sigma_2^2 > \sigma_1^2$

$\alpha = 0.01$

$S_1^2 = 0.3170$

$S_2^2 = 0.1563$



$$F = \frac{0.3170}{0.1563}$$

$$= 2.03$$

$\therefore$ Fail to reject H_0 due to $F = 2.03 < F_{0.01, 11, 15} = 3.73$.

this is because at the level of 0.01 level of significance, we do not have significant evidence to conclude that the variance of the course evaluation scores for female is higher than the male professor.

degree of freedom :

numerator, $n_1 - 1 = 12 - 1 = 11$

denominator, $n_2 - 1 = 16 - 1 = 15$

$F_{0.01, 11, 15} = 3.73$

Question 7

a)

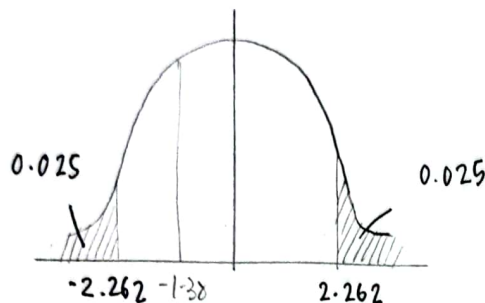
Height mothers	Height daughter	D	D ²
68.0	68.5	-0.5	0.25
60.0	60.0	0	0
61.0	63.5	-2.5	6.25
63.5	67.5	-4	16
69.0	68.0	1	1
64.0	65.5	-1.5	2.25
69.0	69.0	0	0
64.0	68.0	-4	16
63.5	64.5	-1	1
66.0	63.0	3	9
		$\Sigma D =$ -9.5	$\Sigma D^2 =$ 51.75

b)

$$\begin{aligned}\bar{D} &= \frac{\Sigma D}{n} \\ &= \frac{-9.5}{10} \\ &= -0.95\end{aligned}$$

$$\begin{aligned}s_D &= \sqrt{\frac{\Sigma D^2 - \frac{(\Sigma D)^2}{n}}{n-1}} \\ &= \sqrt{\frac{51.75 - \frac{(-9.5)^2}{10}}{10-1}} \\ &= 2.1788 \\ &= 2.18\end{aligned}$$

c) $\alpha = 0.05$ $df = n - 1$ $H_0 : \mu_D = 0$
 $= 10 - 1$ $H_1 : \mu_D \neq 0$
 $= 9$



From table,
critical value, $T = 2.262$

$$t_D = \frac{\bar{D} - \mu_D}{\frac{s_D}{\sqrt{n}}}$$

$$= \frac{-0.95 - 0}{\frac{2.18}{\sqrt{10}}}$$

$$= -1.38$$

$$t_D > t_{0.025}$$

$$-1.38 > -2.262$$

Conclusion :

Fail to reject H_0 .

There is not enough evidence to support the claim that there is no difference in heights between mothers and their first daughters.

Question 8

Using the provided data:

- a. calculate the value for
- e_i
- ,
- $[O_i - e_i]^2 / e_i$
- and
- χ^2

Cell, i	Observed Count, O_i	Expected count, e_i	$[O_i - e_i]^2 / e_i$
1, 1	90	$(200)(200)/430 = 93.023$	0.098
1, 2	80	$(200)(170)/430 = 79.070$	0.011
1, 3	30	$(200)(60)/430 = 27.901$	0.157
2, 1	70	$(150)(200)/430 = 69.761$	0.001
2, 2	60	$(150)(170)/430 = 59.302$	0.008
2, 3	20	$(150)(60)/430 = 20.930$	0.041
3, 1	40	$(80)(200)/430 = 37.209$	0.209
3, 2	30	$(80)(170)/430 = 31.628$	0.084
3, 3	10	$(80)(60)/430 = 11.163$	0.121
$\sum \chi^2$			0.729

Test statistic, $\chi^2 = 0.729$

- b.)
- H_0
- : There is no significant association between customer satisfaction levels and their preferred method of communication.

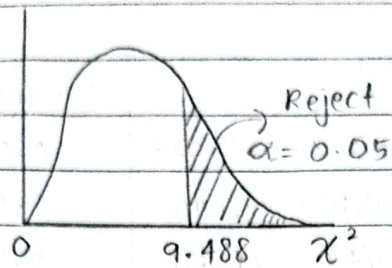
 H_1 : There is a significant association between customer satisfaction levels and their preferred method of communication.

$$\chi^2 = 0.729$$

$$\begin{aligned} \text{degree of freedom (df)} &= (r-1)(c-1) \\ &= (3-1)(3-1) \\ &= 4 \end{aligned}$$

significance level, $\alpha = 0.05$

Critical value,



Decision:

Since $0.729 < 9.488$, we fail to reject the null hypotheses.

Conclusion:

There is no significant association between customer satisfaction levels and their preferred method of communication.